Haoran Sun

B.S. Student in Mathematics, Fudan University
Research Intern, Computational Robotics Lab
Harvard University

Email: haoran_sun@seas.harvard.edu
Homepage: https://haoran1828.github.io/
Phone: 213-651-2363
LinkedIn | Scholar

Education

Fudan University, Shanghai, China
B.S. in Mathematics and Applied Mathematics — 9/2022 – 6/2027 (expected)
School of Mathematical Sciences
Overall GPA: 3.77/4.00
GPA (Fall 2025–Present): 3.95/4.00

University of California, Los Angeles, Los Angeles, CA
Exchange Student in Mathematics — 9/2025 – 12/2025
GPA: 4.00/4.00

Related Courses

UCLA (graduate): MATH 273A Basic Optimization Theory (A+); MATH 275A Probability Theory (A+); MATH 269A Advanced Numerical Analysis (A)

Fudan: Optimization (A); Numerical Solution of Differential Equations (A); Real Analysis (A); Deep Learning (A); Mathematical Modeling (A); Ordinary Differential Equations (A); Stochastic Processes (A); Mathematical Statistics (A)

Research Interests

Numerical Optimization; Semidefinite Programming; Large-Scale Optimization; GPU-Accelerated Methods; Numerical Linear Algebra

I aim to develop *efficient computational algorithms* for large-scale optimization problems, particularly *GPU-accelerated methods*, together with *rigorous theoretical foundations*.

My current research focuses on: (a) developing scalable optimization algorithms that leverage insights from graph theory and numerical linear algebra while harnessing modern computing hardware, particularly GPUs; and (b) investigating the mathematical structure of optimization problems, particularly sparse moment relaxations, to understand when locally extracted atoms can be extended to global minimizers.

Experience

Harvard University, Cambridge, MA
Research Intern, Computational Robotics Lab — 1/2026 – present
Advisor: Prof. Heng Yang

- Established theoretical guarantees for extracting globally extendable atomic candidates from sparse moment relaxations, even when only selected cliques satisfy simplicial regularity. Developed a progressive extraction method based on it and demonstrated its effectiveness on unicycle trajectory optimization.
- Developed spectral deflation for factorization-free matrix filtering, with provable improvements in Newton–Schulz projection accuracy. Integrated it into the Muon optimizer and ADMM for SDP, improving performance in language-model pretraining and large-scale SDP solving.

Lumin Capital, Shanghai, China

Quantitative Research Intern 7/2024 – 9/2024

- Built a text-driven statistical arbitrage strategy: extracted vector features from analyst research reports, trained XGBoost and GRU models to predict forecast revisions and excess returns, and constructed factor signals for backtesting.

Publications

[1] Haoran Sun. Global Extendability of Local Atoms in Sparse Moment Relaxations. arXiv:2610.17371, 2026. (arXiv)

[2] Haoran Sun, Shucheng Kang, and Heng Yang. Spectral Deflation for Factorization-Free Matrix Filtering in Muon and Semidefinite Programming. arXiv:2609.21102, 2026. (arXiv)

Teaching

Teaching Assistant, Calculus, Fudan University Spring 2025

International School of Finance; language of instruction: English

Honors and Awards

William Lowell Putnam Mathematical Competition 2026

Top 120 of all participants; ranked 1st at UCLA (Basil Gordon Prize, $1,000)

Silver Medal, Chinese Mathematical Olympiad (CMO) 2021

First Prize Scholarship (top 5%), Fudan University 2025

First Prize (1st place), East China Mathematical Modelling Competition 2024

Outstanding Student Leader, Fudan University 2023